

Question 1: Colour Representations & Greyscale Conversions (3 pts: 0.5 each)



Colour Image



Greyscale Version 1



Greyscale Version 2

For the colour image, consider the locations at the center of the four squares next to letters (A-D).
For square A, what is the RGB value?

[147, 34, 52]

[254, 254, 254]

[16, 8, 12]

[255, 201, 18]

[112, 111, 58]

How to tell: red component should dominate over other RGB values; [254, 254, 254] is white; cannot be [255, 201, 18], too much green;

For the colour image, consider the locations at the center of the four squares next to letters (A-D).
For square B, what is the normalized RGB value?

[0.63, 0.15, 78]

[0.54, 0.42, 158]

[0.33, 0.33, 254]

[0.44, 0.22, 12]

[0.40, 0.40, 40]

How to tell: yellow mixes equal components red and green with little to no blue; see L1, slide 49 Intensity cannot be 254 (means R, G and B are all saturated); [0.4, 0.4, 40] is possible but intensity is lower, so of the two [0.54, 0.42, 158] is more likely;

For the colour image, consider the locations at the center of the four squares next to letters (A-D).
For square C, what is the V of the HSV value?

[147, 254, 16, 40, 112]

How to tell: definition of V is the max (R, G, B); Location C on a white pixel

For the colour image, consider the locations at the center of the four squares next to letters (A-D).
For square D, what is the intensity?

[78, 158, 254, 12, 40]

How to tell: definition of Intensity is (R+G+B)/3; Location D is on a near-black pixel

Which weighting $[W_R, W_G, W_B]$ was used to convert the colour image to greyscale image version 1?

[0.950, 0.025, 0.025], [0.025, 0.950, 0.025]

[0.025, 0.025, 0.950], [0.299, 0.587, 0.114]

[0.33, 0.33, 0.33]

How to tell: red values tend to saturate

Which weighting $[W_R, W_G, W_B]$ was used to convert the colour image to greyscale image version 2?

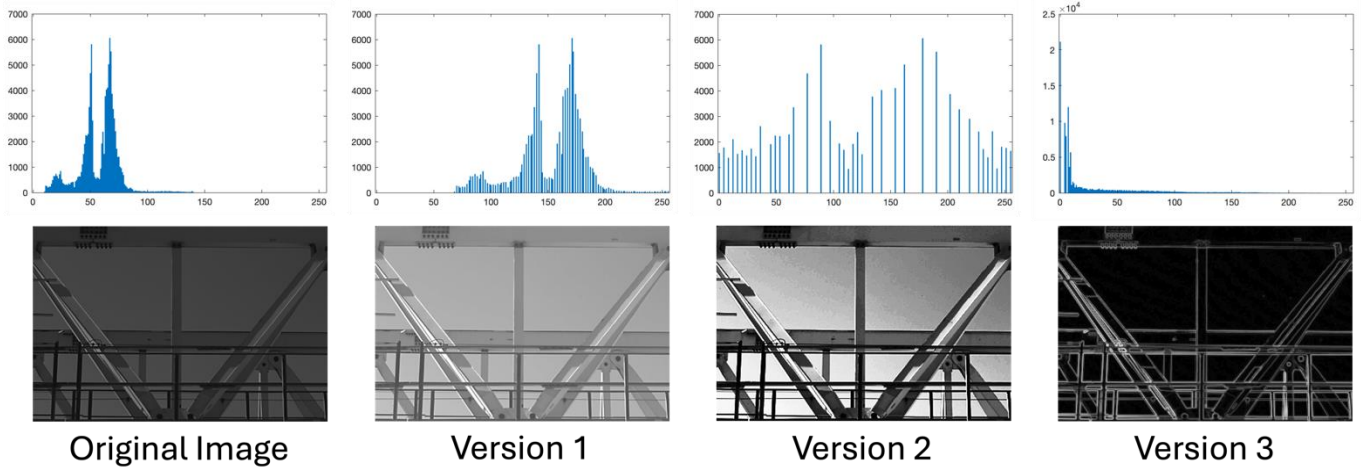
[0.950, 0.025, 0.025], [0.025, 0.950, 0.025]

[0.025, 0.025, 0.950], [0.299, 0.587, 0.114]

[0.33, 0.33, 0.33]

How to tell: red and green values suppressed

Question 2: Image Histograms (4 pts)



Consider the original image on the left and three transformed versions of it. Each image's associated intensity histograms is plotted above. Select the appropriate transformation that leads to each of the three versions.

Version 1

Brightness Adjustment + Intensity Scaling

Gamma Mapping

Histogram Stretching

Histogram Equalization

Image Gradient Magnitude

Canny Edge Detection

Median Filtering

None of the other options

How to tell: Shape of histogram is approximately scaled and shifted.

It is not histogram scaling because the lowest values don't project to 0.

Version 2

Brightness Adjustment + Intensity Scaling

Gamma Mapping

Histogram Stretching

Histogram Equalization

Image Gradient Magnitude

Canny Edge Detection

Median Filtering

None of the other options

How to tell: spacing of the bins and zero-values around it is proportional to the counts; the higher the peaks in the histogram, the larger the zero-counts around it, which exactly correspond to histogram equalization.

Version 3

Brightness Adjustment + Intensity Scaling

Gamma Mapping

Histogram Stretching

Histogram Equalization

Image Gradient Magnitude

Canny Edge Detection

Median Filtering

None of the other options

Canny output is single-pixel, binary image.

Magnitude has range of values.

(Otsu's) Linear operation applied to intensities will lead to the same shift so corresponding threshold will be found in version 1. (ignoring the minor clipping effect). Will also accept no if the explanation emphasizes the clipping effect.

You apply Otsu's method to the images to try and separate the foreground (metal structures) from the background (sky). Will applying Otsu's method to transformed Version 1 lead to the same foreground-background segmentation as the original?

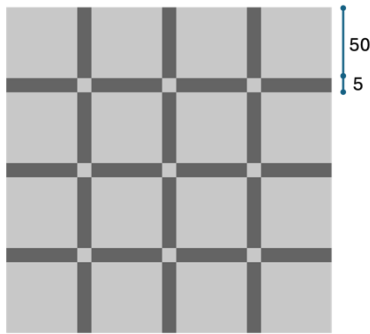
Yes / No / Can't tell

Will applying Otsu's method to transformed Version 2 lead to the same foreground-background segmentation as the original?

Yes / No / Can't tell

Uneven shift of the histogram weights in the equalization operation.

Question 3: Filtering & Segmentation (7 pts)



Original Image

In the original image above, the large squares are 50x50 of intensity 200, while the bars are of height 5, with intensity 100.

What bit-depth is required to encode the original image? (1,2,4,5,6,7,8)

Binary image

Consider convolving the original image with the following filters. Do not pad borders. When applying normalized filters, convert outputs back to integers between 0-255. How many unique intensity values will there be when convolving with a

- (a) a horizontal Sobel filter: **5 unique values**
- (b) a 2D Laplace filter $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$: **5 unique values**
- (c) a normalized Gaussian filter of 3x3: **6 unique values**
- (d) a normalized box filter of 55x55: **1 unique value**

How to solve: for (a) – (c) this question requires finding the number of unique 3x3 summation windows; there are 18 possible unique windows in the input image, but due to symmetries in the kernels, there are less unique values.

(d) the kernel is sized such that it always includes the same number of white and dark pixels.

Consider that the original image is corrupted with noise. Assume that the noise is spatially distributed in a uniformly random manner, but each noise pixel does not have any other noise pixels in its 8-neighbour vicinity. What is the maximum number of unique intensity values in an output image if

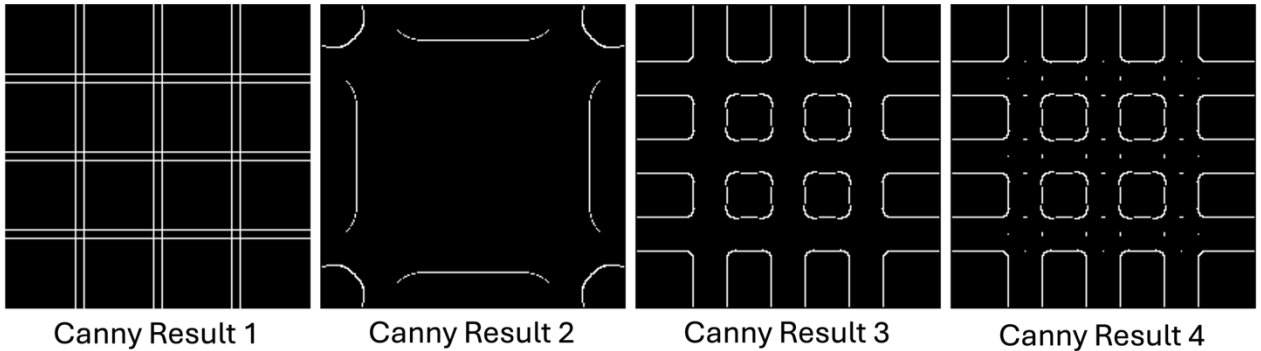
- (e) the noise is salt-and-pepper noise and you apply a 3x3 median filter : **2 unique values**
- (f) the noise is additive Gaussian noise, drawn from a normal distribution $\mathcal{N}(0, 20^2)$:

72 (max # of corrupted pixels post median filtering) + 2 (original 100 / 200) = 74.

To solve question, must understand the sorting operation used in median filtering. Since only 1 pixel within any window is corrupted at any point, for salt and pepper noise, the corrupted pixel (of value 0, 255) will never be the median so the original image will always be recovered.

For Gaussian noise, this is a little harder. The standard deviation of the Gaussian is set s.t. that any resulting value of pixel (0-255) can result. However, any single corrupt pixel <100 or >200 will never be the median. So the unique values must be between the range of 100 and 200. Furthermore, there are only two types of windows which can be corrupted (4 light pixels + 5 dark) or (5 light pixels + 4 dark), this occurs at the corners. There are 72 possible corners.

Question 4: Canny Edge Detection (3 points)



You run Canny edge detection on the original image. Depending on the parameter settings, you get 4 different results shown above. Consider the following pairs of Canny results and specify the parameter settings that lead to the different results.

Canny results 1 and 3:

- The sigma of the Gaussian filter for Result 1 is smaller than Result 3
- The sigma of the Gaussian filter for Result 1 is larger than Result 3
- The gradient magnitude threshold for Result 1 is set lower than Result 3
- The gradient magnitude threshold for Result 1 is set higher than Result 3
- The high threshold of hysteresis thresholding of Result 1 is set lower than Result 3
- The high threshold of hysteresis thresholding of Result 1 is set higher than Result 3
- Result 1 uses gradient interpolation during non-maximum suppression whereas Result 3 uses the nearest pixel approximation

Canny results 2 and 3:

- The sigma of the Gaussian filter for Result 2 is smaller than Result 3
- The sigma of the Gaussian filter for Result 2 is larger than Result 3
- The gradient magnitude threshold for Result 2 is set lower than Result 3
- The gradient magnitude threshold for Result 2 is set higher than Result 3
- The high threshold of hysteresis thresholding of Result 2 is set lower than Result 3
- The high threshold of hysteresis thresholding of Result 2 is set higher than Result 3
- Result 2 uses gradient interpolation during non-maximum suppression whereas Result 3 uses the nearest pixel approximation

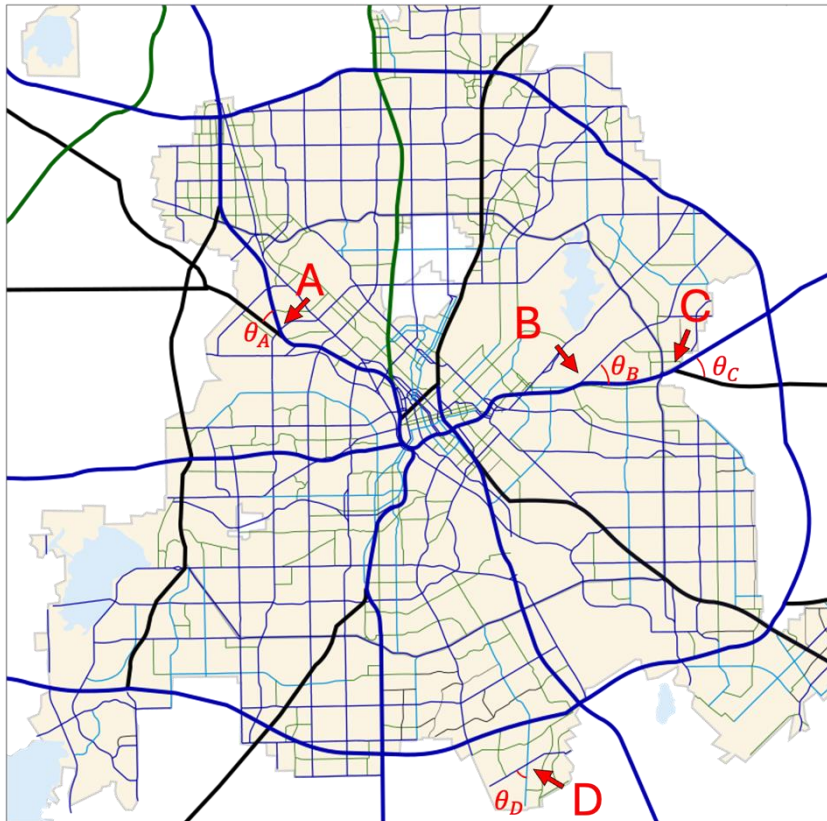
Canny results 3 and 4:

- The sigma of the Gaussian filter for Result 3 is smaller than Result 4
- The sigma of the Gaussian filter for Result 3 is larger than Result 4
- The gradient magnitude threshold for Result 3 is set lower than Result 4
- The gradient magnitude threshold for Result 3 is set higher than Result 4
- The high threshold of hysteresis thresholding of Result 3 is set lower than Result 4
- The high threshold of hysteresis thresholding of Result 3 is set higher than Result 4
- Result 3 uses gradient interpolation during non-maximum suppression whereas Result 4 uses the nearest pixel approximation

How to tell. The position of the found edges give indication on the sigma used for the Gaussian filtering. The larger the sigma, the more the edges get spread. (Recall Lecture 3 slide 19). This relationship governs the differences between Canny Result 1 and 3 as well as 2 and 3. In the most extreme case, the intermediate edges (the middle horizontal and vertical bar) gets swallowed so Canny result 2 can detect only the outer-most edges.

The gradients for small white squares are weaker (after Gaussian filtering) so only a lower threshold picks them up, hence they are present in Canny Result 4 but not in 3. If the Gaussian filtering sigma is very small, this won't be affected, which is why in Canny Result 1, we are able to identify the edges of the small squares cleanly.

Question 5: System Design (6 points)



Dallas Road Map: 1 pixel = 100m

Consider the above road map of the city of Texas, where each pixel represents a real-world distance of 100m. There are three types of roads: highways (thick blue, thick black lines), regular (dark blue thin lines) and minor (light blue, green thin lines). You are tasked with finding all the intersections formed by straight roads of at least 2 km, and the angle of intersection between the two roads, θ , is between 30 to 60 degrees, i.e. $30^\circ \leq \theta < 60^\circ$. Valid intersections Some examples are marked by red arrows in the above map.

- (a) Outline key steps of a method to find the intersections using the Hough transform. State any assumptions made for your solution. (4 points)
- (b) Assume that you wish to find only the 30-60 degree intersections with highways (marked examples A and C) and exclude those in which one (B) or both roads (D) are regular or minor. What changes would you make to your method given in (a)? (1 point)
- (c) Assume that you wish to find the 30-60 degree intersections, excluding intersections where one or both roads are light blue minor roads (marked example D). What changes would you make to your method given in (a). (1 point)

Question 4: System Design (solution)

(a) Method (4 points)

- *(1 points)* Preprocess by converting into edge image with e.g. Canny and exclude edges with insufficient contrast e.g. grey borders lakes, etc.
- *(2 points)* Use standard voting for line detection, with the normal form of parameterization (theta / rho). For all lines found, check in pairwise manner, constraints: (1) if their angles of intersection are within the range of 30-60 degrees (2) if the point of intersection is within the boundaries of the image and (3) the composing lines have sufficient votes.

*(1 points for line voting + some form of finding intersection;
1 point for checking constraints: angles, suf. votes / long enough / within boundary).*

A common mistake here was voting for lines of 30-60 degrees.

I accepted 2/3 constraints to give full marks on the constraints.

Other possible solutions using e.g. generalized Hough transform of the intersection, or direct voting for intersection.

- *(1 point)* Assumptions: the light grey lines, boundaries of the lakes, etc. can be reliably excluded, or any other reasonable assumption wrt the pre-processing or voting procedure (any valid assumption will be accepted)

(b) Canny edge detection, pre-processing, select sufficiently large sigma to ensure that thin line are not detected, so they are not accounted for in the vote (1 point)

- Colour filtering not accepted as solution since dark blue highways and roads
- Only saying set threshold of Canny "higher" not accepted

(c) Colour filtering + fill in with background colour to "erase" the light blue roads (1 point)